

Implementing Scrum in a Distributed Software Development Organization

Hubert Smits - Rally Software Development - hubert@rallydev.com

Guy Pshigoda – BMC Software - guy_pshigoda@bmc.com

Abstract

This paper describes the start of the implementation of Scrum within BMC Software (Identity Management business unit). It describes the project, the project structure and the first 18 months of the project with the impediments and solutions. The specific attention points addressed are around the work with distributed teams (team members of a single team in multiple countries, as well as whole teams in different countries). The size of the project at hand is significant: 50 people are involved, and the initial duration is estimated at 18 months.

1. Introduction: The Start of the Agile Implementation

The BMC Identity Management organization is part of the larger BMC group. At the time IdM's agile implementation started in December 2005, another BMC team, the BMC OnePatrol group, had collected good experiences with its implementation of Scrum*. Based on these experiences, the IdM management team decided to adopt Scrum as the method of choice for the implementation of the next major version of the Identity Management suite.

1.1 The Approach for Moving to Agile

The primary principles the IdM group applied for its move to agile were to: 1) grow an implementation from a small initiative to a full program and 2) add a program roadmap and release planning level to the Scrum framework to create a long-term view of events.

This approach meant that three teams in Tel Aviv started working in a Scrum fashion to develop their development environment. They created a product backlog, ran the Sprint planning, and worked in time-boxes. The results were encouraging, the involved teams were enthusiastic, and they delivered working

software. In parallel, the product roadmap was developed by the product owners and the product backlog was completed.

The steps used to transform the organization into agile practices have also been described previously in two papers – The CIO's Playbook for Adopting the Scrum Methodology[†] and Multi-level Planning in Agile Projects[‡].

1.2 The Project and the Structure of the Teams

The project started with four teams. Two of the teams had participants from multiple locations (Tel Aviv, France and Florida), and the other two teams had participants in Tel Aviv only. The teams were created around product functions, and the motivation to have people from more than one location participate in a single team was based on the desire to create a knowledge transfer between the locations.

The initial teams were made up primarily of developers. There was a lack of testers, tech writers and support (build) people, as they were tied up in another project. The mitigation for this situation was to have the developers take on these additional roles.

Product owners were in a team of their own, not truly in the delivery teams. The product owner team consisted of five product owners and a manager located across Israel (4), France (1) and the US (1). One architect was assigned to the project, who was also outside the delivery teams and not reporting into the management structure that made up the delivery and product owner team.

The lead product owner created a concise description of the program, with a logical product roadmap, which was accepted by all involved people. A simplified structure of the roadmap looked like this:

- Release 1 – proof of architecture through minimal business functionality, reducing technology risks

* Israel Gat, "How BMC Is Scaling Agile Development", Agile2006 conference

[†] Ken Schwaber, Hubert Smits, Dean Leffingwell - 2005

[‡] Hubert Smits – HICSS 2007 conference

- Release 2 – working demo, 60% of full functionality, usable to collect customer feedback
- Release 3 – near full business functionality, suite ready for deployment on selected customer sites
- Release 4 – full functionality (including documentation, installer, data migration) available for deployment at selected customer sites
- Release 5 – full functionality, SDK, data migration, product shippable to customers

The product owner team worked hard to create a product backlog out of the existing documentation. This process proved to be difficult, and resulted in an incomplete backlog with many hundreds of features and a large variation in feature size ranging from hours to weeks of work.

1.3 First Release Planning

With the experience in the teams, the completed product roadmap and the available product backlog in hand, the whole team was invited for the first release planning. All team members were flown into Tel Aviv to participate in training and planning. The structure of the training was received enthusiastically by all participants, and all participants used the time together well to create a team spirit, bond and work through many questions and issues.

Release planning took a full day, with all four teams participating and a total of 35 participants in the room. We used the following agenda:

- Opening, introductions, agenda review
- Product vision – formulated by the VP of engineering, the lead product owner and the lead architect
- Release time-boxes, number of Sprints, available capacity in the teams
- Discussion and agreement of ‘done’ for a feature
- Activities to be planned for in the release (new features, support, next Sprint preparation, other activities)
- Moving features into Sprints, on a team-by-team basis
- Group review of the team-based release plans to find dependencies, risks, and mitigations
- Retrospective for the day

The day completed successfully, with a release plan that was accepted by everyone involved. This plan was the basis for the Sprint planning sessions of the four teams, which commenced the following day. Sprint planning was not formally facilitated. Instead, the teams used their experience of the earlier Sprints and

were all successful in the completion of the plans. They agreed to have the individual plans reviewed by the four ScrumMasters, the development manager and the architect. This approach worked well to uncover any dependencies and risks.

1.4 Early Experiences and Early Impediments

The early impediments had little to do with Scrum methods and were mainly related to the unavailability of people and resources. The lack of testers was flagged early and repeatedly by all teams. Problems with missing hardware were quickly resolved.

Feedback from the involved teams was mainly positive. People liked the increased level of self-organization, and the ability to plan and own their own work. Product owners went through significant efforts to create a product backlog that had the right level of detail and contained both business and architecture features. The translation of architectural requirements into backlog features proved to be difficult; neither architects nor product owners had worked in this fashion before.

2. First Process Evaluation

Well into the second release, management decided to leverage the idea of working agile. The decision was made to change the scope of the project due to business needs brought by product management. The idea was to take most of what the team had done so far and add to it some new directions and market trends. Since change is welcomed in agile projects, all teams continued working to complete the current release. At the same time, management and product management worked on the preparation of the new release.

The major concerns from project management from a Scrum perspective were the new team structure (more Scrum teams in more locations) and how to get the new team members up to speed with both Scrum and the project.

While the scope of the project was reviewed and extended, the steering committee started to review the Scrum implementation. This initiative was led by the VP of engineering with the goal to evaluate many aspects of the Scrum implementation thus far and set guidelines for the next phase.

First, all steering committee members (VP Engineering, Product Management Manager, CTO, Senior Development Manager, Project Manager, and Support Director) presented their findings. Then the members began interviewing different individuals from each

area of the project: QA manager and testers, tech writers, developers from different teams and sites, ScrumMasters, architects, and product owners. The committee focused on the following areas.

2.1 Scrum: Is It the Right Solution?

Once the evaluation process began, it seemed natural to ask the question, “The project is big, complex and faced many obstacles so far - is Scrum really the right methodology of choice?”

I remember during last year’s conference I asked Ken Schwaber, “Are there projects that are not meant to be Scrum projects?” To make a long story short, he said: Don’t use Scrum if you want to ignore your impediments and if you don’t want to know that your project may fail in an early stage.

The same thing happened in this project. We faced many issues and concerns that could have been attributed to using Scrum. Together with our agile coach, we understood that if it wasn’t for Scrum, we most likely would have ended up in the same situation several months later after spending more money and having fewer alternatives.

Product Owner Involvement and Interaction with the Teams: We have five product owners that are part of the project; some of them are responsible for more than one component and in more than one location. Adding to that, the horizontal product owner is located in the US, which makes effective day-to-day communication even harder.

Since product owners have a crucial role in Scrum at all levels, we wanted to ensure involvement of the product owner for each team as well as managing the backlog and preparing the next planning phase.

Because the project backlog is feature-driven rather than story-driven, it makes the planning, development, testing and review processes harder. We looked for a way to change that.

The Architects Role: One of the main challenges of creating the new suite in our environment is to look at the different teams and remember that we are talking about one product. Part of that task fell with the architects. They had to ensure a solid infrastructure, and they should be working closely with the product owners. Some misunderstandings and responsibility conflicts appeared between the architects and product owners, especially around architecture requirements - how should they appear in the backlog and who owns them?

Product Quality, QA Involvement, and Assuring Homogeneity: As mentioned previously, the lack of QA resources was a known impediment from the beginning, and had a known solution. The main question was how to handle the overall testing of the suite, create and test end-to-end scenarios, govern the quality of the different teams and set standards for all of the teams. There are testers in the delivery teams, but who are they reporting to? Should it require an organizational change? If so, what is the role of the QA manager?

Documentation – Per Team and the General Product Docs: Shared development services from a non-agile project that are now part of the delivery team in a Scrum project raised lots of questions, and documentation is still an open issue. The first impediment was regarding planning. How can we develop, document and then test the documentation in the same Sprint? Lots of discussions were and still are ongoing about how to handle this problem. One idea was to work in a document-driven development approach. Another idea was to have the technical writers lag behind the rest of the team by one sprint.

What about documents that are horizontal and not part of any of the teams such as concept guide and release notes? Who is the owner? Who will test them?

Planning and Execution: Last but definitely not least is planning and execution. For each planning session, we invested lots of effort bringing people over from all locations, so it was crucial to be highly efficient.

We tried to identify the key success factors for having a good planning session. The first key factor was to have everybody, or at least as many people as possible, attend the planning in one location.

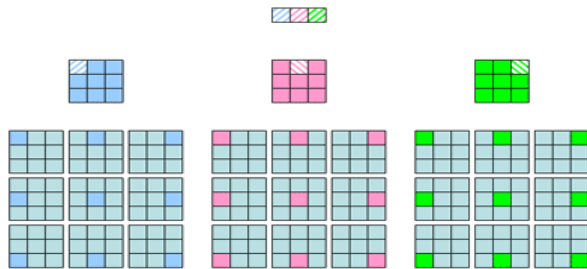
A second key factor was determining how long before the planning session the backlog should be ready? What is “ready”? For execution, what are the ways to help the delivery teams execute better and to meet their commitments?

3. Solutions and Recommendations

After few good weeks of hard work gathering all the information from the different groups, as well as meeting with representatives from the BMC Patrol group, we presented the conclusions and recommendations to the project steering committee, then to the entire R&D group.

3.1 Nested Scrum

With 14 delivery teams, having 14 ScrumMasters reporting into one “Scrum of Scrums” in a 15-minute daily meeting just looked too big, certainly when adding the fact that the teams resided in four different time zones. We had to scale up one level and have multiple, daily Scrum of Scrum meetings, led by a senior ScrumMaster. The senior ScrumMaster reports to the project Scrums of Scrums of Scrums, which was led by the senior development manager. In this way, we kept the daily meeting principal and allowed for a more reasonable forum in each one of the meetings. The biggest challenge was to keep the flow of information all the way to the top and back again to the delivery teams. This is the structure we chose:



3.2 Validation Team

From the evaluation process, it was clear that we needed to emphasize that the product is a suite, making sure that we looked at all of the sets of components that are developed by the different teams as one big product. We created the Suite Validation Team with the following mission: “To ensure that the Suite, as a whole product, satisfies the market needs and provides the desired BMC IdM experience along with the required documentation and the best quality.” The Suite Validation Team acted as a Scrum team like all others led by a ScrumMaster.

The team defines typical customer configurations and use-case scenarios. They create the test book for all end-to-end scenarios, perform installation testing, and oversee the overall customer experience working with the new Identity Management suite.

We invested a lot of thought about who the team members should be. On one hand we needed people who would have this vision and can present to customers. On the other hand they have to be “working” or active members, not only consultants. The ScrumMaster was chosen to be the suite QA manager. Other members were: Documentation Lead, Tech Leads, Architects, Product Owners, Testers and representatives

from the field. The team will report daily to the Scrum of Scrums of Scrums.

Planning, Planning, Planning... In the very early stage of the evaluation process, we identified that we do face some issues with planning. This may sound naïve in that the teams should just be able to plan better. However, we found that we must consciously modify some of the actions and steps that we were taking. This resulted in more efficient planning sessions, better execution on a single sprint, and, even more importantly, more predictability of the content for the entire coming release.

One of the first items on the list was to start using the “Rally tool,” which gave us the ability to manage all the requirements for such a big project with multiple teams and geographies. We previously planned on a flipchart using sticky notes, which seems impossible when your team is in more than one location.

Next in line was modifying the backlog. As mentioned before, we needed the backlog “ready” on time. To us, “ready” meant creating one big backlog for the entire projects (not one backlog per team) and having it in the right priority order (1-n). Then we defined that in order to prepare for release planning, the teams needed time to work on the backlog, learn it, and assign T-shirt sizes for each feature. We agreed on a reasonable planning time of two weeks.

On one side we had product owners who are customer-oriented and on the other side we had a delivery team that in some ways talks a little bit differently. In those two weeks before the planning started, the architects acted as a liaison between the two groups, translating the requirements into the required technical language.

Once the teams have finished their planning, we set aside two hours for a unified planning meeting in which each ScrumMaster presented the plan for the Sprint and the dependencies that were identified during the planning. The goal of the unified planning meeting was to overcome possible problems due to dependencies between the different teams in each sprint. In doing so, we engaged all ScrumMasters and all product owners to the entire plan of the sprint and have an end-to-end vision.

Integration: We have already discussed the new suite validation team, but we also established two more cross-team Scrum teams to handle integration issues- the Documentation team and the Integration and Automation team. The Documentation team manages books and documents that are cross-team documents and are not a part of any one team. The team also gov-

erns and sets best practices for the tech writers in each of the delivery teams.

The mission of the Integration and Automation team is to produce a stable environment for tests and coordinate cross-suite technical activities. The goal is simply to create an integration environment for all supported platforms and run and review the nightly build and tests for all the teams.

4. About the organizations and the authors

4.1 BMC Software (Identity Management Business Unit)

BMC Software, Inc. is a leading provider of enterprise management solutions that empower companies to manage their IT infrastructure from business perspective, BMC software's Business Service Management (BSM) strategy and in particular its Identity Management solutions are aligned with business needs to ensure that the challenges of today's IT environments are successfully addressed.

Guy Pshigoda, is the project manager of the Identity Management suite, as well as a certified Scrum-Master. Coaching and leading the Scrum implementation in the Identity Management Business Unit.

4.2 Rally Software

From an initial pilot project to enterprise rollout, Rally helps companies succeed with Agile software development. Rally's family of Agile life cycle management products give teams the visibility and collaboration needed to deliver high-value software in rapid iterations, and its world-renowned coaching services help mentor teams to create internal Agile experts. For more information, visit www.rallydev.com.

With more than 15 years of software project management and IT expertise, Hubert Smits has helped hundreds of software team members successfully transition dozens of projects to Agile and Lean practices.

* Mountain Goat Software website -
http://www.mountaingoatsoftware.com/scrum_team